1 Introduction

In this report our main aim is to study whether observable characteristics of the stop, the suspect and the policing context are associated with the probability that an NYPD stop results in an arrest.

Our primary research question is therefore:

> Which characteristics of the suspect, stop circumstances, and police context are associated with the probability that an NYPD stop results in an arrest?

We also consider two secondary research questions. First, we investigate whether the probability of arrest varies across New York City boroughs after accounting for observed stop characteristics. Second, we examine whether there is any residual variation in arrest probability across police command codes after adjusting for suspect and stop characteristics.

The dataset we considered to answer those questions is the Stop, Question and Frisk Data of the New York Police Department for year 2025 which, at the time of this study, is freely available at the following link: `https://www.nyc.gov/site/nypd/stats/reports-analysis/stopfrisk.page`

Among all the variables available in the dataset, we decided to retain a reduced set of covariates that are directly related to our research questions, while avoiding unnecessary dimensionality at the preprocessing stage and in the initial model-building phase. Depending on the results of the preliminary analysis, we may later expand the set of variables in order to capture additional information.

The selected variables include:

1. **Outcome variable**

   The main response variable in this report is `SUSPECT_ARRESTED_FLAG`, a binary variable indicating whether the suspect was arrested after the stop.

2. **Suspect characteristics**

   We retain variables describing the main observable characteristics of the suspect: `SUSPECT_REPORTED_AGE`, `SUSPECT_SEX`, `SUSPECT_RACE_DESCRIPTION`, `SUSPECT_HEIGHT`, and `SUSPECT_WEIGHT`. These variables may help explain whether individual characteristics are associated with different arrest probabilities.

3. **Stop context and dynamics**

   To describe the context and nature of the stop, we include `STOP_WAS_INITIATED`, `SUSPECTED_CRIME_DESCRIPTION`, `OBSERVED_DURATION_MINUTES`, `STOP_DURATION_MINUTES`, and `WEAPON_FOUND_FLAG`. These variables capture how the stop began, the suspected offence, the duration of the observation and of the stop itself, and whether a weapon was found.

4. **Police actions during the stop**

   We include `FRISKED_FLAG` and `SEARCHED_FLAG` as indicators of police actions taken during the stop. These variables may capture the degree of intrusiveness or intensity of the interaction, distinguishing between a frisk and a more intrusive search of clothing, pockets, or belongings.

5. **Temporal variables**

   In order to account for possible temporal patterns, we retain `STOP_FRISK_DATE` and `STOP_FRISK_TIME`. These variables allow us to explore whether the probability of arrest varies across different times or days.

6. **Spatial information**

   To study spatial variation, we include `STOP_LOCATION_BORO_NAME`, which identifies the borough in which the stop took place.

7. **Police unit identifier**

   Finally, we retain ISSUING_OFFICER_COMMAND_CODE, which identifies the police command responsible for the stop. This variable is particularly important because it allows for potential hierarchical modelling of residual variation across police units and may help capture heterogeneity in arrest patterns across commands.

2 Data preprocessing

After selecting the subset of variables from the original dataset that were directly relevant to the research questions we checked for the presence of duplicates using the variable STOP_ID. As no duplicates were found, it was removed from the analysis dataset.

Missing values were encoded as (null) in the original file. These were converted to NA, and observations with missing values in the selected variables were removed, yielding a complete-case dataset for the initial analysis.

Categorical variables with a limited number of levels were encoded as factors.

For SUSPECTED_CRIME_DESCRIPTION, levels with frequency below 0.1% were grouped into a single OTHER category in order to reduce sparsity.

Finally, the numeric variables were inspected through summary statistics, simple graphical checks and correlation matrix to spot redundancies. The two duration variables, OBSERVED_DURATION_MINUTES and STOP_DURATION_MINUTES, were highly right-skewed, so we used the transformation

$$\log(1 + x).$$

This reduced the influence of extreme values and produced variables on a more suitable scale for modelling, while also allowing zero values to be retained.

3 Modelling phase

In the following section we discuss about modelling specification. The resulting models were fitted using INLA.

3.1 Model 1

As a first step, we fit a Bayesian logistic regression model with fixed effects only. The aim of this model is to provide a simple baseline specification for the main research question, and understand at a base level which observable characteristics of the suspect and of the stop are associated with the probability of arrest. More complex sources of variation, such as heterogeneity across police commands, will be taken into consideration later, when we will specify more complex models.

For each stop $i = 1, \dots, n$, let

$$Y_i = \begin{cases} 1, & \text{if the stop results in an arrest,} \\ 0, & \text{otherwise.} \end{cases}$$

We assume

$$Y_i \mid p_i \sim \text{Bernoulli}(p_i),$$

where $p_i = \Pr(Y_i = 1)$ is the probability that stop i results in an arrest. The model uses the logit link,

$$\text{logit}(p_i) = \eta_i,$$

with linear predictor

$$\begin{aligned}\eta_i = \beta_0 + \beta_1 \widetilde{\mathrm{Age}}_i + \beta_2 \log(1 + \widetilde{\mathrm{StopDuration}}_i) + \beta_3 \,\mathrm{Sex}_i \\ + \sum_r \gamma_r \,\mathrm{Race}_{ir} + \sum_s \delta_s \,\mathrm{StopInitiated}_{is} + \sum_c \alpha_c \,\mathrm{CrimeType}_{ic} \\ + \beta_4 \,\mathrm{Frisked}_i + \beta_5 \,\mathrm{Searched}_i + \beta_6 \,\mathrm{WeaponFound}_i + \sum_b \lambda_b \,\mathrm{Borough}_{ib}.\end{aligned}$$

The continuous covariates are centred, while categorical covariates are included through dummy variables (one hot encoding) with one reference category omitted in each case.

Prior specification In the first model, we assigned a separate prior to the intercept and to the regression coefficients. This reflects their different roles in the logistic model: after centring the continuous covariates, the intercept represents the baseline log-odds of arrest for the reference profile, whereas the remaining coefficients quantify changes in log-odds associated with the covariates. We therefore used a weaker prior for the intercept and more regularising gaussian priors for the regression coefficients, specifically:

$$\beta_0 \sim \mathcal{N}(0, 10^2),$$

and

$$\beta_j \sim \mathcal{N}(0, 5^2), \qquad j = 1, \ldots, p.$$

Assumptions The main assumptions of Model 1 are the following:

- conditional on the covariates, the observations are independent;
- the response variable follows a Bernoulli distribution;
- covariate effects are additive on the log-odds scale;

Below we report a brief summary of the INLA output.

Coefficient	Mean	Lower95	Upper95
(Intercept)	-4.536	-4.699	-4.373
SEARCHED_FLAGY	2.905	2.814	2.995
SUSPECTED_CRIME_DESCRIPTIONFORCIBLE TOUCHING	2.808	2.093	3.522
SUSPECTED_CRIME_DESCRIPTIONPETIT LARCENY	2.530	2.360	2.701
SUSPECTED_CRIME_DESCRIPTIONRAPE	2.442	1.331	3.553
SUSPECTED_CRIME_DESCRIPTIONCRIMINAL MISCHIEF	2.383	2.087	2.678

Table 3.1 shows the five coefficients with the largest posterior mean in absolute value. In this logistic regression model, coefficients are interpreted on the *log-odds* scale. For categorical predictors, each coefficient represents the change in the log-odds of arrest relative to the chosen reference category, holding the remaining covariates fixed. For continuous predictors, the coefficient represents the change in log-odds associated with a one-unit increase in the covariate.

The posterior mean of the intercept is -4.53. Since the continuous covariates were centred, this can be interpreted as the baseline log-odds of arrest for the reference profile, evaluated at the mean values of the continuous predictors. Converting this quantity to probability gives

$$\frac{\exp(-4.53)}{1 + \exp(-4.53)} \approx 0.011,$$

so the baseline probability of arrest for the reference profile is approximately 1.1%.

Among the largest positive coefficients, the strongest effect is associated with SEARCHED_FLAGY. This suggests that, conditional on the other covariates, stops involving a search are associated with substantially higher odds of arrest than stops without a search. Large positive coefficients also appear for some categories of SUSPECTED_CRIME_DESCRIPTION, in particular FORCIBLE TOUCHING, PETIT LARCENY, RAPE, and CRIMINAL MISCHIEF. These coefficients are all interpreted relative to the reference crime category used in the model, which is "CPW" (Criminal possession of a Weapon). These suspected offences are then associated with higher odds of arrest than the reference category.

The 95% credible intervals reported in the table are entirely above zero for these coefficients, which indicates a positive association with arrest on the log-odds scale.

3.2 Model 2: temporal and spatial extension

The first model only included fixed effects and therefore assumed that the probability of arrest did not exhibit any additional temporal or spatial structure. In this context, such an assumption is likely to be too restrictive. Since the stop-and-frisk data contains both temporal information and broad spatial information, it is reasonable to allow the model to capture patterns that may vary over time and across locations. In particular, the probability of arrest may change over time because of seasonal effects, differences between daytime and nighttime stops, or shifts in policing intensity. Similarly, even after controlling for the observed characteristics of the stop and of the suspect, some residual heterogeneity may still persist across boroughs. For this reason, the second model extends the baseline specification by introducing structured temporal effects together with a borough-level random effect to account for residual spatial variation.

Let Y_i denote the arrest indicator for stop i, with

$$Y_i \mid p_i \sim \text{Bernoulli}(p_i), \qquad i = 1, \dots, n.$$

Using the logit link, the model is specified as

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} + f_{\text{date}}(d_i) + f_{\text{time}}(h_i) + u_{b(i)},$$

where:

- β_0 is the intercept;
- x_{ij} denotes the observed covariates already included in Model 1 (excluding Borough that now is modelled as a random effect);
- $f_{\text{date}}(d_i)$ is a smooth temporal effect associated with the calendar date of stop i;
- $f_{\text{time}}(h_i)$ is a smooth effect associated with the time of day of stop i;
- $u_{b(i)}$ is a borough-specific random intercept, where $b(i)$ indicates the borough in which stop i took place.

The two temporal components are introduced to capture distinct types of variation. The calendar date effect describes gradual changes in arrest probability across the observation period, whereas the time of day effect captures systematic within-day variation.

A natural choice for the calendar date component is a first order random walk:

$$f_{\text{date}}(d_t) - f_{\text{date}}(d_{t-1}) \sim \mathcal{N}(0, \tau_{\text{date}}^{-1}),$$

which encourages nearby dates to have similar effects while still allowing for smooth evolution over time.

The time of the day component is instead modeled as a cyclic latent effect. More specifically, it can be represented through a cyclic first order random walk:

$$f_{\text{time}}(h_t) - f_{\text{time}}(h_{t-1}) \sim \mathcal{N}(0, \tau_{\text{time}}^{-1}), \qquad f_{\text{time}}(h_1) - f_{\text{time}}(h_H) \sim \mathcal{N}(0, \tau_{\text{time}}^{-1}),$$

where H denotes the last point in the discretized 24 hour cycle. In this way, nearby times are assigned similar effects and the pattern is constrained to join smoothly at midnight. This specification is important because clock time is periodic: for example, 23:55 and 00:05 are close in practice, even if they are far apart on a linear scale.

The spatial component is modeled at the borough level through a random intercept:

$$u_b \sim \mathcal{N}(0, \tau_{\text{boro}}^{-1}), \qquad b = 1, \ldots, B.$$

In this model, borough is treated as a random effect in order to account for residual area-level heterogeneity in arrest probabilities. More specifically, it is included to absorb unobserved spatial variation that may still remain after controlling for the observed characteristics of the stop.

Assumptions The additional assumptions of Model 2 with respect to model 1 are the following:

1. the covariates affect the arrest probability through the logit link;
2. the latent effect for calendar date evolves smoothly over adjacent dates, while the effect of time of day varies smoothly over adjacent times and is modeled cyclically over the 24-hour period;
3. the borough-specific effects are exchangeable realizations from a common Gaussian distribution;

As in the baseline model, weakly informative Gaussian priors are assigned to the fixed effects:

$$\beta_0 \sim \mathcal{N}(0, 100), \qquad \beta_j \sim \mathcal{N}(0, 25), \quad j = 1, \ldots, p.$$

For the latent components, we specify penalized complexity priors through the condition

$$\Pr(\sigma > 1) = 0.01,$$

where $\sigma = \tau^{-1/2}$ denotes the standard deviation of the corresponding latent effect. This implies regularization toward a simpler benchmark, namely the absence of temporal or spatial variation, unless the data provide sufficient evidence for more structure.

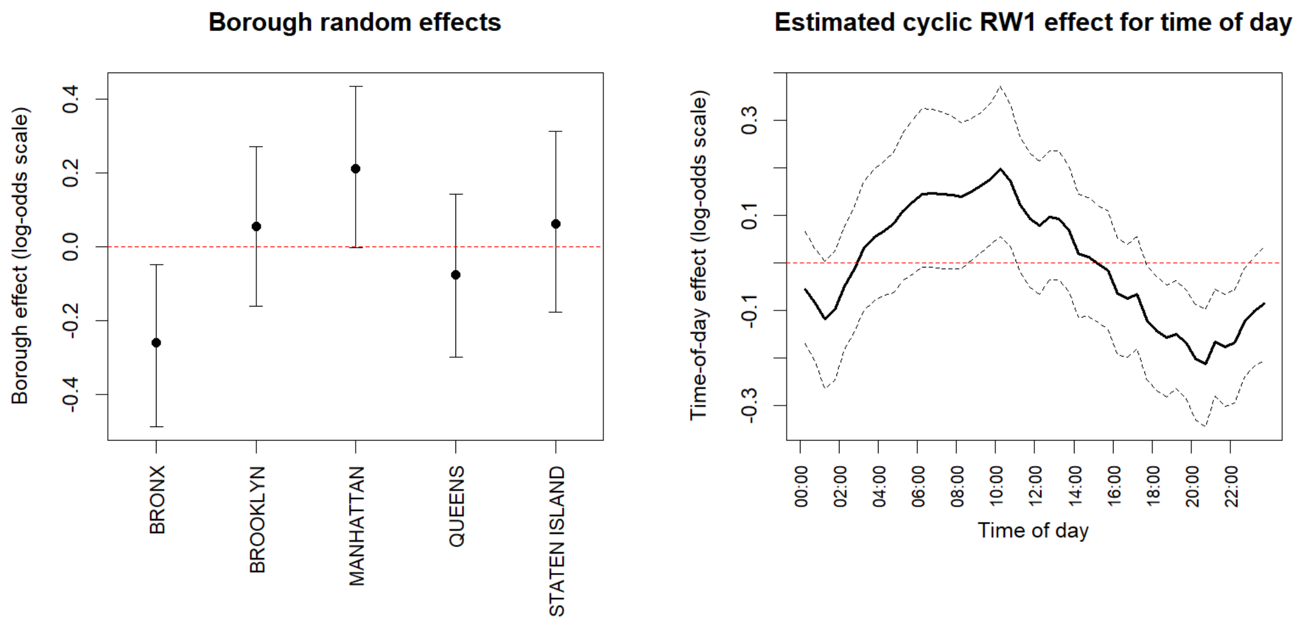


Figure 3.2 provides a graphical summary of the latent components introduced in Model 2. The left panel shows the posterior mean and 95% credible interval of the borough-level random effects on the log-odds scale. Since these effects are centered around zero, positive values indicate boroughs

with a higher arrest probability than the overall average, after controlling for the observed stop and suspect characteristics, while negative values indicate boroughs with a lower arrest probability.

The figure suggests that some residual heterogeneity across boroughs remains even after adjustment for the observed covariates. In particular, Manhattan displays the largest positive posterior mean, whereas the Bronx shows a clearly negative effect. Brooklyn and Staten Island also have positive posterior means, while Queens is slightly below zero. At the same time, most credible intervals still overlap zero, which suggests that the evidence for borough differences is not equally strong across all areas. The clearest signal is for the Bronx, whose interval lies almost entirely below zero, pointing to lower residual log-odds of arrest relative to the citywide average. Overall, the graph suggests that borough level variation is present, but its magnitude is moderate once the other predictors are taken into account.

The right panel shows the estimated cyclic random walk effect for time of day. This effect is also represented on the log-odds scale, with the dashed lines indicating the 95% credible band. The pattern is not flat, which supports the choice of including a structured within-day temporal component rather than treating time as irrelevant. The estimated effect tends to increase from the early morning hours, reaches its highest values in the first half of the day, and then declines during the afternoon and evening, becoming negative in the late evening and night. This result suggests that stops taking place at different times of day may be associated with different arrest probabilities. The 95% credible bands, however, remain fairly wide over part of the day and cross the zero log-odds level several times.

3.3 Model 3: hierarchical extension

The second model already improved on the baseline specification by introducing structured temporal effects and a borough-level random effect. However, an additional source of dependence may still be present in the data, since stops are naturally grouped by issuing officer command. In practice, stops associated with the same command may resemble each other more than stops associated with different commands, even after accounting for the observed covariates, the temporal structure, and the borough-level heterogeneity. If this grouping is ignored, part of the residual variability may remain unexplained.

For this reason, the third model extends Model 2 by adding a hierarchical component based on `ISSUING_OFFICER_COMMAND_CODE`.Therefore, in this specification, rather than estimating a completely separate parameter for each command, command-specific effects are treated as draws from a common distribution. This makes it possible to account for heterogeneity across commands while at the same time borrowing strength across groups. Such an approach is particularly appropriate when some commands are represented by relatively few observations, as in our case.

Let Y_i denote the arrest indicator for stop i, with

$$Y_i \mid p_i \sim \text{Bernoulli}(p_i), \qquad i = 1, \ldots, n.$$

Using the logit link, the model is specified as

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \sum_{j=1}^{p} \beta_j x_{ij} + f_{\text{date}}(d_i) + f_{\text{time}}(h_i) + u_{b(i)} + v_{c(i)},$$

where:

- $v_{c(i)}$ is the command-specific random intercept, where $c(i)$ identifies the issuing officer command associated with stop i.

The additional hierarchical component is specified as

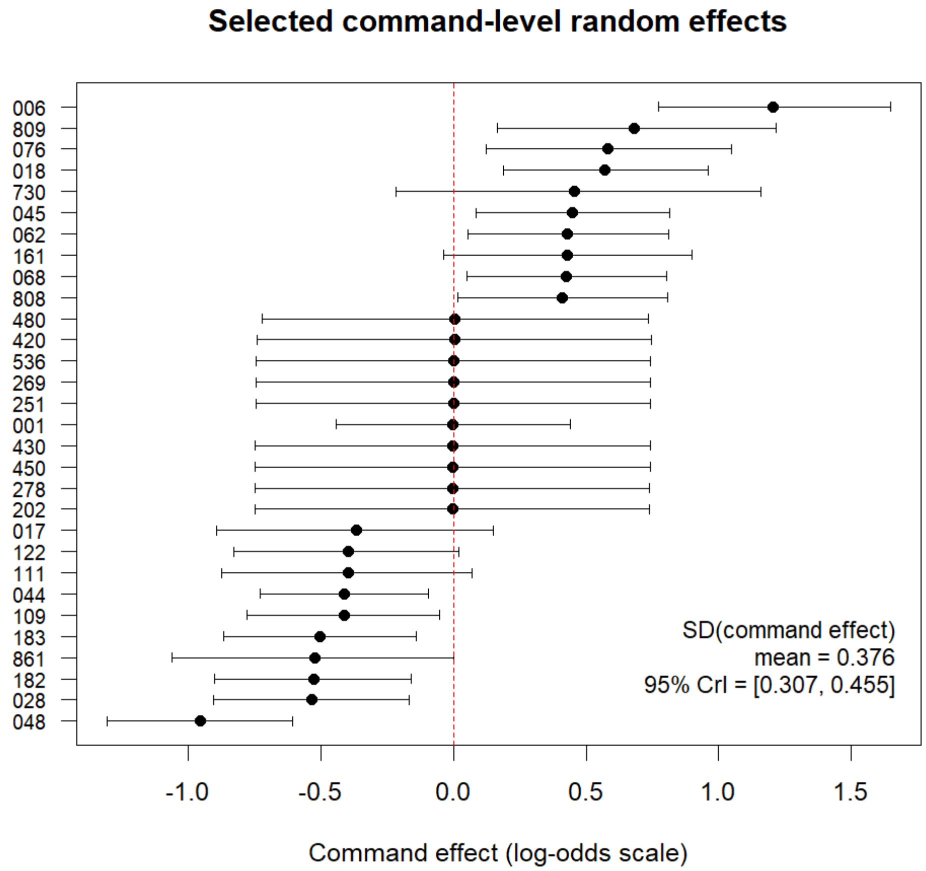
$$v_c \sim \mathcal{N}(0, \tau_{\text{cmd}}^{-1}), \qquad c = 1, \ldots, C.$$

This means that each command is allowed to have its own deviation from the overall log-odds of arrest, while these deviations are assumed exchangeable and centered around zero. The prior assumptions remain the same as in the previous model. In particular, weakly informative Gaussian priors are assigned to the fixed effects, while penalized complexity priors are specified for the precision parameters of all latent random effects, including the additional command-level effect.

Assumptions The additional assumption in Model 3 relative to Model 2 concerns the hierarchical structure introduced in the model. More precisely:

1. the command-specific effects are exchangeable realizations from a common Gaussian distribution;

The plot 4 below suggests that there is some residual variation across police command codes, but not an overwhelmingly large one. Most command-level effects remain fairly close to zero, and many credible intervals still cross the zero line, which means that for a large number of commands the evidence of a clearly positive or negative deviation is limited. Still, a few commands do stand out, showing effects that are more clearly above or below zero. This means that, after adjusting for the observed characteristics of the suspect and of the stop, as well as for temporal and borough-level variation, some residual heterogeneity across commands remains. The posterior mean of the command-level standard deviation is 0.376, with 95% credible interval 0.307–0.455, which points to a real but moderate grouping effect.



4 Model Comparison

Looking across the three specifications, the general picture is quite consistent. The main fixed effects remain almost the same from one model to the next, both in sign and in magnitude, as we can see in the table below. In particular, the strongest positive associations are always linked to the type of suspected offence and to whether the suspect was searched.

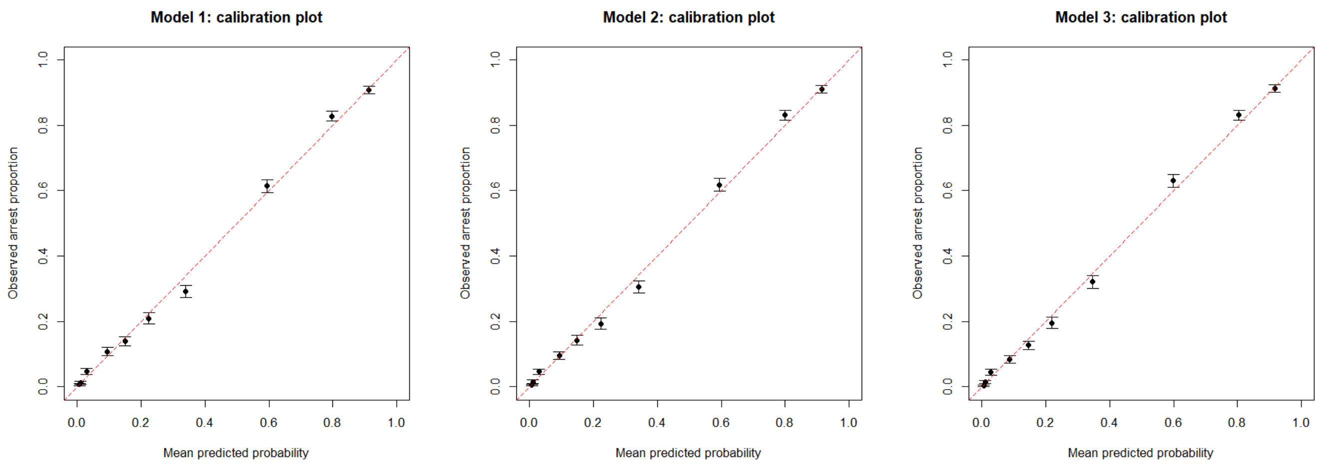
Comparison of top fixed effects across models

Coefficient	Mean M1	Low 95% M1	Up 95% M1	Mean M2	Low 95% M2	Up 95% M2	Mean M3	Low 95% M3	Up 95% M3
(Intercept)	-4.536	-4.699	-4.373	-4.157	-4.408	-3.905	-4.130	-4.386	-3.869
SEARCHED_FLAGY	2.905	2.814	2.995	2.915	2.824	3.007	2.975	2.880	3.069
SUSPECTED_CRIME_DESCRIPTIONFORCIBLE TOUCHING	2.808	2.093	3.522	2.781	2.062	3.501	2.780	2.052	3.508
SUSPECTED_CRIME_DESCRIPTIONPETIT LARCENY	2.530	2.360	2.701	2.496	2.322	2.669	2.475	2.296	2.654
SUSPECTED_CRIME_DESCRIPTIONRAPE	2.442	1.331	3.553	2.384	1.269	3.498	2.423	1.317	3.529
SUSPECTED_CRIME_DESCRIPTIONCRIMINAL MISCHIEF	2.383	2.087	2.678	2.361	2.063	2.659	2.336	2.033	2.639

What changes across the three models is mainly the way residual structure is handled. Model 2 adds temporal effects and a borough-level random effect, while Model 3 further introduces a command-level hierarchical component. These additions do not really overturn the interpretation of the fixed effects, but they do allow the model to capture sources of variation that are left unexplained in the simpler specification.

This is also reflected in the model comparison criteria. The DIC goes from 16125.48 in Model 1 to 16068.20 in Model 2, and then drops further to 15849.61 in Model 3. The same pattern is observed for the WAIC, which decreases from 16124.96 to 16067.60 and then to 15848.71. The NLSCPO moves in the same direction, from 8062.49 to 8033.81 and finally to 7924.39. So, while the move from Model 1 to Model 2 gives a moderate improvement, the hierarchical extension in Model 3 leads to a more visible gain in fit.

As a final descriptive check, we also looked at calibration plots based on grouped fitted probabilities for the three models.



These plots are useful here because they give a simple way to compare predicted arrest probabilities with the corresponding observed frequencies. All three models show a fairly reasonable alignment with the 45 degree line, so at least at an aggregate level the fitted probabilities seem coherent with the data. That said, these plots are based on the same data used to fit the models, so they should be read as an internal diagnostic rather than as a true out of sample validation.

5 Conclusion

The analysis suggests that the probability of arrest is mainly related to the suspected offence and to some key features of the stop, especially whether the suspect was searched. These effects remain fairly stable across the different model specifications, which supports the robustness of the main findings. The results also point to limited variation across boroughs, but somewhat clearer differences across police command codes. Overall, Model 3 appears to provide the best fit to the data.